

D_report — Full Technical Report

Titanic Survival Prediction — Technical Report *Mini Project 01 | Foundations of ML | Zuucrew*

1. Preprocessing Summary

The Titanic dataset contained 891 passengers across 12 features. Four columns were removed immediately — PassengerId, Name, and Ticket carried no predictive signal, while Cabin had 77% missing values making imputation unreliable. The remaining 7 features were retained for modelling.

Missing values in Age (177 instances) were imputed using the median rather than the mean, as age distribution is right-skewed and the median is more robust to outliers. The two missing Embarked values were filled with the mode — the most frequent port of embarkation.

Categorical variables were encoded numerically. Sex was mapped to binary (male=0, female=1) and Embarked to ordinal integers (C=0, Q=1, S=2). All features were standardised using z-score normalisation computed on training data only and applied to both splits — preventing data leakage. A stratified 80/20 train/test split was created using random seed 42, maintaining the original 38.38% survival rate in both subsets.

2. Key Results Comparison Table

Model	Accuracy	Precision	Recall	F1	AUC
LR No Reg (Scratch)	0.7989	0.7619	0.6957	0.7273	0.8438
LR L1 (Scratch)	0.7989	0.7619	0.6957	0.7273	0.8442
LR L2 (Scratch)	0.7989	0.7619	0.6957	0.7273	0.8441
Bagging (Scratch)	0.7765	0.7377	0.6522	0.6923	0.7059
AdaBoost (Scratch)	0.8045	0.7576	0.7246	0.7407	0.8208
LR No Reg (Sklearn)	0.7989	0.7797	0.6667	0.7188	0.8518
LR L1 (Sklearn)	0.7989	0.7797	0.6667	0.7188	0.8516
LR L2 (Sklearn)	0.7989	0.7797	0.6667	0.7188	0.8518
Bagging (Sklearn)	0.7765	0.7377	0.6522	0.6923	0.7534
Random Forest	0.8101	0.8431	0.6232	0.7167	0.8464
AdaBoost (Sklearn)	0.7821	0.7500	0.6522	0.6977	0.8249
Gradient Boosting	0.7989	0.7895	0.6522	0.7143	0.8203
HistGradient Boosting	0.7877	0.7719	0.6377	0.6984	0.8196

3. Logistic Regression Variants Analysis

All three Logistic Regression variants — no regularisation, L1, and L2 — produced identical accuracy of 79.89% and AUC scores clustered tightly around 0.844. This convergence indicates the Titanic dataset is not prone to overfitting at this feature dimensionality, making regularisation largely redundant here.

The learned weight analysis revealed Sex as the dominant predictor with a coefficient of 0.99, confirming the historical "women and children first" evacuation policy. Pclass followed at -0.55, reflecting the survival disadvantage faced by third-class passengers. Fare contributed positively at 0.25, correlated with class. Parch showed near-zero weight at -0.02, indicating family size had minimal independent survival influence once class and gender were accounted for.

L1 regularisation did not zero out any weights, confirming all seven features carry genuine predictive signal. L2 produced marginally smaller weights across all features without eliminating any, consistent with its behaviour of proportional shrinkage rather than sparsity.

4. Bagging vs Boosting Performance

Bagging with Decision Stumps produced the weakest results — 77.65% accuracy and AUC of 0.71 in the from-scratch implementation. Decision Stumps are inherently weak base learners, limited to a single binary split per model. While bagging reduces variance effectively, it cannot compensate for the high bias of stumps on a multi-feature dataset. Random Forest, which extends bagging with feature randomness and deeper trees, significantly outperformed basic bagging — achieving 81.01% accuracy and 0.85 precision, the highest precision of all models tested.

AdaBoost demonstrated the clearest advantage of sequential ensemble learning. The from-scratch implementation achieved 80.45% accuracy and 0.8208 AUC, outperforming the sklearn AdaBoost at 78.21%. This result validates the correctness of the manual implementation and highlights how adaptive weight adjustment effectively focuses learning on misclassified passengers. Gradient Boosting and HistGradient Boosting performed comparably to Logistic Regression at approximately 79% accuracy, suggesting diminishing returns on this relatively simple dataset.

5. Performance Conclusion

Random Forest emerged as the best overall model with 81.01% accuracy and the highest precision of 0.8431, making it the most reliable at correctly identifying survivors. The from-scratch AdaBoost produced the best recall of 0.7246 among ensemble methods, capturing the highest proportion of actual survivors.

The near-identical performance between scratch and sklearn Logistic Regression implementations validates the mathematical correctness of the from-scratch gradient descent. The primary performance ceiling across all models appears to be dataset size — 891 samples limits the complexity of patterns learnable — rather than algorithmic choice.

6. Future Recommendations

Feature Engineering — Creating interaction terms such as family size ($\text{SibSp} + \text{Parch} + 1$) and extracting titles from passenger names (Mr, Mrs, Miss, Master) would provide richer signal without additional data collection and could push accuracy above 83%.

Hyperparameter Optimisation — A systematic grid search over Random Forest depth, estimator count, and minimum samples per leaf using cross-validation would likely yield a further 1–2% accuracy improvement over the default configuration used here.

XGBoost or LightGBM — Replacing HistGradientBoosting with XGBoost or LightGBM would introduce second-order gradient optimisation, better regularisation control, and built-in handling of class imbalance — directly addressing the 38/62 survival split that constrained recall across all models.